
Can 3D JEPA Segment? Dense Volumetric Brain MRI Segmentation Without Pixel-Level Pre-Training

Han Riman *

Department of Computer Science
Master’s Thesis Research

Abstract

Joint-Embedding Predictive Architectures (JEPA) are formulated to learn abstract semantic representations by predicting missing regions in latent feature space rather than reconstructing raw pixel or voxel intensities. This design philosophy creates a fundamental question for volumetric medical image segmentation: *can an architecture trained without voxel-level reconstruction perform dense, per-voxel volumetric segmentation?* In this work, we investigate this question through a controlled empirical study on multi-modal brain MRI segmentation (BraTS 2024 Glioma). We evaluate a teacher-free single-encoder 3D JEPA equipped with Variance-Invariance-Sketching Regularization (**3D VisReg**) and a multi-scale 3D Vision Transformer Feature Pyramid Network (ViT-FPN) decoder against fully supervised volumetric convolutional baselines: a 3D Residual UNet (MONAI) and a 3D DynUNet with multi-scale deep supervision. On held-out test volumes ($N = 271$), 3D VisReg JEPA demonstrates that latent-space pre-training successfully enables dense segmentation, achieving $79.71 \pm 14.34\%$ Dice and 11.04 ± 14.36 mm HD95. However, it trails supervised CNNs on full training data ($86.42 \pm 12.06\%$ for ResUNet; $87.60 \pm 11.93\%$ for DynUNet) due to the coarse spatial granularity of 16^3 patch tokenization. Under severe annotation scarcity (1%–5% labels), self-supervised pre-training provides substantial utility, outperforming standard 3D Residual UNet (47.21% vs. 3.98% at 1% labels; 58.16% vs. 4.70% at 5%) by preventing the low-data background collapse observed in unregularized CNNs under class imbalance and short training schedules, though deeply supervised DynUNet remains superior across all tiers (56.65% at 1%). In inference throughput, VisReg JEPA runs $2.05\times$ faster than DynUNet (114.57 ms vs. 235.22 ms per 128^3 volume), while standard Residual UNet is fastest (75.54 ms). Out-of-distribution evaluations reveal critical boundary conditions: while resilient to mild additive Rician noise, 3D JEPA exhibits pronounced vulnerability to smooth B_1 radiofrequency field inhomogeneities (59.98% Dice) and missing contrast-enhanced sequences (37.28% on T1c-only). These findings demonstrate that while 3D JEPA can segment without voxel reconstruction and offers benefits under extreme label scarcity, it incurs measurable trade-offs in boundary precision, full-supervision ceiling, and scanner artifact robustness.

1 Introduction

Automated volumetric segmentation of pathological brain lesions in multi-parametric Magnetic Resonance Imaging (MRI)—such as gliomas imaged via co-registered T1, post-contrast T1 (T1c), T2, and Fluid-Attenuated Inversion Recovery (FLAIR) acquisitions—is a cornerstone of neuro-oncology diagnosis, surgical boundary planning, and conformal radiotherapy targeting [Baid et al., 2024, Menze et al., 2014]. Fully supervised 3D convolutional architectures, notably

*Correspondence to: hanriman@master.thesis

3D U-Net [Ronneberger et al., 2015, Çiçek et al., 2016], V-Net [Milletari et al., 2016], and 3D nnU-Net [Isensee et al., 2021], have long established the performance standard in volumetric medical image segmentation. Their success is commonly attributed to high-resolution isotropic spatial feature hierarchies and dense horizontal skip connections, which route low-level edge, gradient, and texture cues directly from early encoder layers into the volumetric decoder, ensuring precise spatial localization of anatomical boundaries.

However, training supervised 3D deep models from random initialization requires hundreds of densely contoured 3D volumetric masks delineated across dozens of contiguous axial slices by expert neuroradiologists. Manual voxel contouring is prohibitively expensive, labor-intensive, and subject to marked inter-observer discordance [Menze et al., 2014]. Self-Supervised Learning (SSL) offers an appealing paradigm to alleviate annotation bottlenecks by pre-training representation backbones on unannotated imaging cohorts prior to downstream supervised adaptation.

Generative self-supervised approaches, such as 3D Masked Autoencoders (3D MAE) [He et al., 2022, Feichtenhofer et al., 2022], learn representations by masking spatial voxels or patches and training an expansive decoder to reconstruct raw voxel intensities in image space. In multi-modal brain MRI, however, high-frequency voxel variance is heavily contaminated by non-biological acquisition artifacts: radiofrequency thermal noise (characterized by non-central Rician distributions [Gudbjartsson and Patz, 1995]), scanner-dependent B_1 magnetic field inhomogeneities, and microstructural tissue speckle. Devoting model capacity to 3D voxel-level reconstruction forces the network to allocate expressive resources to uninformative acquisition noise rather than underlying anatomical morphology and pathological lesion boundaries.

The 3D JEPA Paradigm and the Voxel Paradox To overcome the limitations of pixel and voxel reconstruction, Joint-Embedding Predictive Architectures (JEPA) [Assran et al., 2023, LeCun, 2022] operate strictly within latent feature space. Instead of reconstructing masked voxels, an online 3D context encoder processes visible volumetric tokens, and a lightweight predictor network is trained to forecast the latent representations of masked 3D target blocks. By omitting a pixel-level decoder entirely, JEPA is designed to learn abstract semantic representations and spatial contextual relationships while discarding high-frequency noise.

This foundational design choice leads directly to a central research question and an apparent scientific paradox:

Can 3D architectures designed explicitly to avoid voxel-level reconstruction perform dense, per-voxel volumetric segmentation?

At first glance, the objectives appear fundamentally incompatible. Dense volumetric segmentation requires voxel-accurate delineation of infiltrative tumor boundaries across 3D space, whereas JEPA’s objective explicitly incentivizes the encoder to discard high-frequency spatial variation in favor of abstract semantic representations. If low-level spatial details are discarded during pre-training, can the learned representations support downstream volumetric boundary delineation?

Scope and Contributions In this work, we investigate this question through an empirical study comparing 3D Joint-Embedding Predictive Architectures against standard supervised 3D convolutional baselines in multi-modal volumetric brain MRI segmentation. Rather than relying on early dual-encoder JEPA variants that require heuristic momentum teachers, we evaluate **3D VisReg JEPA** [Wu et al., 2026], a single-encoder formulation that prevents latent representation collapse through decoupled optimal transport feature regularization. We pair this encoder with a hierarchical multi-scale 3D Feature Pyramid Network (FPN) decoder to bridge the gap between latent embeddings and dense spatial predictions.

Our contributions are threefold:

1. **Feasibility and Inductive Bias Trade-offs in 3D JEPA:** We demonstrate that a self-supervised 3D JEPA pre-trained purely in latent embedding space without voxel reconstruction learns representations capable of dense volumetric segmentation, reaching $79.71 \pm 14.34\%$ Dice and 11.04 ± 14.36 mm HD95 on held-out test volumes ($N = 271$). We quantify the performance gap relative to supervised 3D CNNs (86.42% ResUNet, 87.60% DynUNet), identifying the structural trade-off between patch tokenization and fine boundary delineation.
2. **Mitigation of Low-Data Background Collapse:** We systematically evaluate volumetric label efficiency across six annotation tiers (1% to 100%). 3D VisReg JEPA substantially outperforms standard 3D Residual UNet under extreme data scarcity (47.21% vs. 3.98% at 1% labels; 58.16% vs. 4.70% at 5%), confirming that self-supervised latent pre-training provides structural initialization that prevents the background collapse afflicting unregularized CNNs under severe class imbalance and limited iteration budgets. However, deeply supervised DynUNet demonstrates that multi-scale architectural supervision also prevents collapse (56.65% at 1%).
3. **Empirical Boundary Conditions, Geometry, and Latency Profiles:** We analyze the representation geometry of single-encoder 3D JEPA (erank = 27.29 on 384-dimensional features, centered cosine similarity 0.0062) and profile inference latency, showing that VisReg JEPA runs $2.05\times$ faster than deeply supervised DynUNet (114.57 ms vs. 235.22 ms), while standard Residual UNet remains fastest (75.54 ms). Crucially, stress tests reveal key vulnerabilities: while robust to mild additive Rician noise, JEPA degrades significantly under macroscopic B_1 field bias (59.98%) and missing contrast sequences (37.28%).

2 Background: Progression of the JEPA Family

Self-supervised non-contrastive learning in latent space faces a fundamental degenerate failure mode: *latent representation collapse*, where the encoder maps all input patches to a constant vector, driving prediction loss to zero without learning informative representations [Bardes et al., 2022, Zbontar et al., 2021]. The evolution of the JEPA family reflects progressive advances in preventing this collapse.

3D I-JEPA: Dual-Encoder Architecture with EMA Teacher The foundational Image-JEPA (I-JEPA) [Assran et al., 2023] prevents representation collapse by maintaining two separate parameter sets: an online context encoder E_θ updated via gradient descent, and a target teacher encoder $E_{\bar{\theta}}$ updated via an Exponential Moving Average (EMA) of online weights:

$$\bar{\theta}_t \leftarrow m\bar{\theta}_{t-1} + (1 - m)\theta_t, \quad m \in [0.996, 1.0] \quad (1)$$

Combined with stop-gradient operators on target representations, the slow evolution of the teacher prevents instantaneous representational collapse [Grill et al., 2020]. However, in 3D volumetric medical imaging, the dual-encoder EMA framework introduces acute drawbacks:

1. **GPU Memory Footprint:** Storing an identical copy of the 3D vision backbone doubles parameter memory during pre-training, severely restricting 3D subvolume batch sizes on memory-constrained GPUs.
2. **Heuristic Scheduling:** Training stability depends heavily on empirical cosine scheduling of teacher momentum $m_t = m_{\text{end}} - (m_{\text{end}} - m_{\text{start}}) \cdot \frac{1}{2}(1 + \cos(\pi t/T_{\text{anneal}}))$ ($m \in [0.996, 1.0]$); improper schedules can induce representation destabilization or asymptotic collapse.
3. **Implicit Mechanism:** Rather than optimizing an explicit objective that penalizes low representation rank or collapsed feature geometry, non-collapse under EMA updating is an emergent optimization phenomenon.

LeJEPA (SIGReg): Teacher-Free Sketched Gaussian Regularization To eliminate the EMA teacher, LeJEPA [Balestrierio and LeCun, 2025] introduced Sketched Isotropic Gaussian Regularization (**SIGReg**), enabling single-encoder JEPA training. By the Cramér-Wold theorem [Cramér and Wold, 1936], a multivariate distribution on $\mathbb{R}^D$ is uniquely determined by the family of its 1D marginal projections onto the unit hypersphere $\mathbb{S}^{D-1}$. SIGReg samples random 1D projections and penalizes the discrepancy between the empirical characteristic function (ECF) of projected features and the standard normal characteristic function via the Epps–Pulley test statistic [Epps and Pulley, 1983]. While SIGReg removes the momentum teacher, it couples representation location, coordinate scale, and higher-order distribution shape into a single non-parametric test statistic, which can complicate optimization dynamics when 3D features require simultaneous scaling and de-correlation.

3D VisReg: Decoupled Optimal Transport Regularization Variance-Invariance-Sketching Regularization (**VisReg**) [Wu et al., 2026] refines teacher-free JEPA pre-training by formally decoupling three geometric degrees of freedom in representation space: (1) batch centering, (2) coordinate scale regularization, and (3) distributional shape regularization via 1D Sliced-Wasserstein Distance (SWD) [Bonneel et al., 2015, Kolouri et al., 2019, Villani, 2009].

Rationale for Focusing on 3D VisReg JEPA In this study, we investigate 3D VisReg JEPA as the representative member of the JEPA family for three concrete reasons:

- **Teacher-Free Single-Encoder Simplicity:** It eliminates the memory overhead and momentum hyperparameter tuning of I-JEPA, making full 3D volumetric pre-training practical on standard accelerators.
- **Principled Geometric Decoupling:** By separating coordinate variance anchoring from non-Gaussian shape penalties, VisReg prevents opposing gradient dynamics between variance expansion and distribution matching.
- **Decoupled Representation Space:** By applying non-collapse regularization to an auxiliary lower-dimensional projector space ($d_p = 128$), the primary encoder feature manifold ($D = 384$) remains unconstrained, allowing it to represent complex anatomical patterns without being forced into an isotropic distribution.

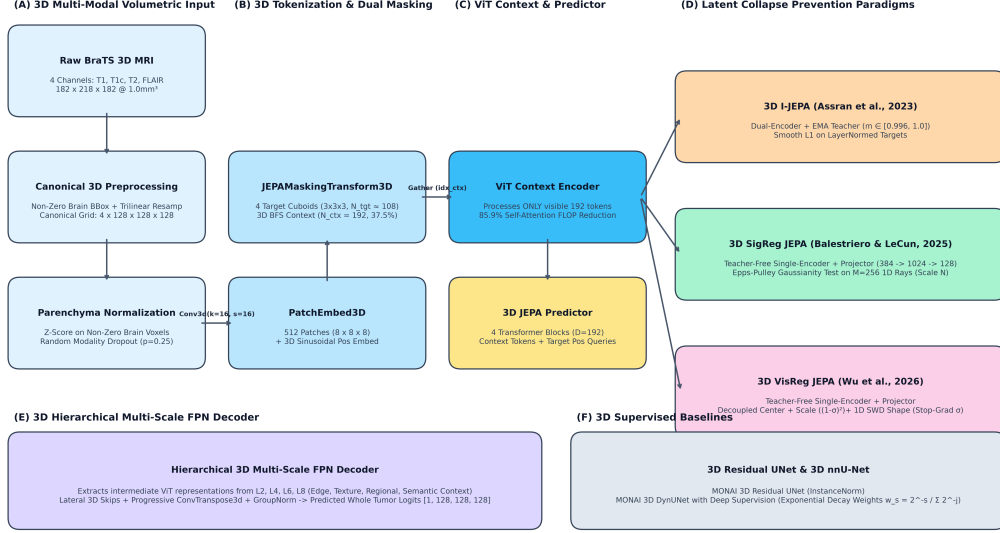


Figure 1: **System Architecture Overview.** (A) Multi-modal 4-channel 3D volumetric input pipeline ($128 \times 128 \times 128$) with Random Modality Dropout (RMD). (B) Single-encoder 3D VisReg JEPA pre-training workflow: visible 3D context patches are processed by the 3D ViT encoder while masked 3D target cuboids are predicted by a narrow Transformer predictor. Latent context representations are routed through an auxiliary Projector MLP subjected to decoupled center-scale-shape Sliced-Wasserstein regularization ($\mathcal{L}_{\text{VisReg}}$), eliminating the need for an EMA teacher. (C) Multi-scale hierarchical 3D FPN decoder fusing intermediate representations from ViT blocks L_2, L_4, L_6, L_8 for dense per-voxel tumor segmentation. (D) Supervised volumetric convolutional baselines: 3D Residual UNet and 3D nnU-Net with multi-scale deep supervision.

3 Methodology

Figure 1 illustrates our end-to-end framework, contrasting single-encoder 3D VisReg JEPA pre-training, our multi-scale 3D ViT-FPN segmentation decoder, and supervised volumetric convolutional baselines.

3.1 Multi-Modal 3D Volumetric Processing and Preprocessing

Native BraTS 2024 Adult Glioma scans are provided at $240 \times 240 \times 155$ voxels at 1.0 mm^3 isotropic resolution across four co-registered sequences: T1, post-contrast T1 (T1c), T2, and FLAIR. To focus computational resources on intracranial tissue and standardize inputs, we extract the 3D bounding box containing all non-zero brain voxels and resample the cropped volume to a canonical grid of $H = W = D = 128$ voxels using trilinear interpolation for images and nearest-neighbor interpolation for ground-truth masks. Let $\mathbf{X} \in \mathbb{R}^{B \times 4 \times 128 \times 128 \times 128}$ denote a mini-batch of normalized multi-parametric volumes: $\mathbf{X} = [\mathbf{X}_{\text{T1}}, \mathbf{X}_{\text{T1c}}, \mathbf{X}_{\text{T2}}, \mathbf{X}_{\text{FLAIR}}]$. The downstream segmentation target is the binary Whole Tumor (WT) mask ($\text{mask} > 0$), which combines all annotated BraTS 2024 lesion sub-compartments (enhancing tissue, non-enhancing tumor core, peritumoral edema, and resection cavity) into a single binary foreground target.

Volume-Level Non-Zero Voxel Standardization To avoid slice-wise intensity inconsistencies and preserve 3D gradient continuity, each volume is independently Z-score standardized across its non-zero intracranial parenchyma:

$$\hat{\mathbf{X}}_c(v) = \frac{\mathbf{X}_c(v) - \mu_{c,p}}{\sigma_{c,p} + \epsilon}, \quad \forall v \in \Omega_{\text{fg}} = \{u \in \mathbb{R}^3 \mid \mathbf{X}_c(u) > 0\} \quad (2)$$

where $\mu_{c,p}$ and $\sigma_{c,p}$ are the empirical mean and standard deviation of channel c within the intracranial brain mask Ω_{fg} of patient volume p , and $\epsilon = 10^{-8}$. Voxels outside the brain mask are set to zero.

Random Modality Dropout (RMD) In clinical neuro-oncology, complete four-sequence protocols are frequently compromised by patient motion, acquisition time limits, or contraindications to gadolinium contrast. To instill intrinsic robustness to missing sequences, volumes undergo Random Modality Dropout during training: each sequence channel is independently zeroed out with probability $p_{\text{drop}} = 0.25$ [Havaei et al., 2016]. A deterministic fallback ensures that if all four channels are selected for dropout, one channel is randomly preserved, preventing empty inputs while maintaining reproducibility.

Patient-Level Stratified Partitioning To prevent data contamination, dataset partitioning is enforced at the patient volume directory level ($\mathcal{P}_{\text{train}} \cap \mathcal{P}_{\text{val}} = \emptyset$ and $\mathcal{P}_{\text{train}} \cap \mathcal{P}_{\text{test}} = \emptyset$ across 1,808 BraTS GLI volumes). Cases (identified by unique volume directories, e.g., BraTS-GLI-00005-100) are stratified into tumor volume quartiles before split assignment (70% train, 15% val, 15% test; 1,266 train, 271 validation, and 271 held-out test volumes), ensuring that lesion volume distributions remain consistent across splits. Fractional low-data subsets (1% to 50%) inherit this quartile stratification.

3.2 3D VisReg JEPA Formulation

3D Vision Transformer Backbone The encoder E_θ partitions a 3D volume into non-overlapping $16 \times 16 \times 16$ volumetric patches, yielding a spatial grid of $N_{\text{vox}} = (128/16)^3 = 8 \times 8 \times 8 = 512$ patch tokens. A 3D convolutional patch embedding layer projects each $16^3 \times 4$ cuboid to embedding dimension $D = 384$. Importantly, each token corresponds to a substantial spatial span of $16 \times 16 \times 16$ voxels in the canonical grid, creating an intrinsic structural bottleneck for high-frequency spatial localization that the downstream decoder must resolve. To incorporate 3D spatial geometry, separable 3D sinusoidal coordinate positional embeddings are added:

$$\mathbf{E}_{\text{pos}}^{\text{3D}}(x, y, z) = [\mathbf{e}_{\sin/\cos}(x); \mathbf{e}_{\sin/\cos}(y); \mathbf{e}_{\sin/\cos}(z)] \in \mathbb{R}^{1 \times 512 \times 384} \quad (3)$$

The token sequence passes through $L = 8$ Pre-Layer Normalization Transformer blocks with residual skip connections, $h = 6$ attention heads (projection dimension $d_k = 64$), and MLP expansion ratio 4 ($D_{\text{ff}} = 1536$) with GELU activations.

Stochastic 3D Patch Masking and Predictor During pre-training, the 3D patch grid is partitioned into context and target sets:

1. $K = 4$ disjoint 3D target cuboids of size $3 \times 3 \times 3$ patches each ($N_{\text{tgt}} \approx 27$ patches per block, total $N_{\text{tgt,tot}} \approx 108$ patches) are sampled across the 3D grid with aspect-ratio constraints $\alpha \in [0.75, 1.5]$: $\mathcal{I}_{\text{tgt}} = \bigcup_{k=1}^K \mathcal{I}_{\text{tgt},k}$.
2. Unmasked candidate patches are assembled using a 3D 26-connected Breadth-First Search (BFS) starting from a random seed to form a spatially contiguous 3D context cluster $\mathcal{I}_{\text{ctx}}$ of $N_{\text{ctx}} = 192$ patches (37.5% of the volume).

Only context patches $\mathcal{I}_{\text{ctx}}$ pass through the online encoder E_θ , producing context representations $\mathbf{Z}_{\text{ctx}} \in \mathbb{R}^{B \times N_{\text{ctx}} \times 384}$. Target representations $\mathbf{Y}_{\text{tgt}} = E_\theta(\mathbf{X})[:, \mathcal{I}_{\text{tgt}}, :]$ are generated by forwarding the full volume through the same online encoder under `torch.no_grad()`.

A narrow 4-block Transformer predictor P_ϕ ($D_{\text{pred}} = 192$, $h = 6$) receives the projected context tokens concatenated with learnable target mask token queries $\mathbf{e}_{\text{mask}} + \mathbf{P}_{\text{tgt}}$, predicting the target representations:

$$\hat{\mathbf{Y}}_{\text{tgt}}^{(k)} = \text{Linear}_{192 \rightarrow 384} \left(\text{LN} \left(P_\phi \left(\left[(\text{Linear}(\mathbf{Z}_{\text{ctx}}) + \mathbf{P}_{\text{ctx}}); (\mathbf{e}_{\text{mask}} + \mathbf{P}_{\text{tgt}}^{(k)}) \right] \right)[:, N_{\text{ctx}} :, :] \right) \right) \quad (4)$$

The prediction loss is computed via Smooth L_1 (Huber) loss against asymmetrically Layer-Normalized target representations:

$$\mathcal{L}_{\text{pred}}(\theta, \phi) = \frac{1}{K} \sum_{k=1}^K \text{Smooth}_{L_1} \left(\hat{\mathbf{Y}}_{\text{tgt}}^{(k)}, \text{LayerNorm}(\mathbf{Y}_{\text{tgt}}^{(k)}) \right) \quad (5)$$

Decoupled Center-Scale-Shape Regularization To isolate non-collapse regularization from spatial representations, context tokens are mapped through a 2-layer Projector MLP: $g_\psi : \mathbb{R}^{384} \rightarrow \mathbb{R}^{1024} \rightarrow \mathbb{R}^{d_p}$ ($d_p = 128$). The projected tokens $\mathbf{Z} \in \mathbb{R}^{T \times d_p}$ ($T = B \cdot N_{\text{ctx}}$) undergo decoupled optimal transport regularization:

1. **Center Regularization** ($\mathcal{L}_{\text{center}}$): Mean feature vectors $\boldsymbol{\mu}_z = \frac{1}{T} \sum_{j=1}^T \mathbf{z}_j$ are penalized to enforce zero-centered representations:

$$\mathcal{L}_{\text{center}}(\mathbf{Z}) = \frac{1}{d_p} \|\boldsymbol{\mu}_z\|_2^2 \quad (6)$$

Centered features are then defined as $\mathbf{z}_{c,i} = \mathbf{z}_i - \boldsymbol{\mu}_z$.

2. **Coordinate Scale Regularization** ($\mathcal{L}_{\text{scale}}$): Channel standard deviations $\sigma_d = \sqrt{\frac{1}{T} \sum_i z_{c,i,d}^2} + \epsilon$ are anchored to unit variance via a two-sided squared penalty:

$$\mathcal{L}_{\text{scale}}(\mathbf{Z}) = \frac{1}{d_p} \sum_{d=1}^{d_p} (1.0 - \sigma_d)^2 \quad (7)$$

3. **Feature Standardization**: Tokens are standardized with a stop-gradient operator on the scale:

$$\mathbf{z}_{s,i,d} = \frac{z_{c,i,d}}{\text{sg}(\sigma_d) + \epsilon} \quad (8)$$

ensuring $\mathbb{E}[\mathbf{z}_s] = \mathbf{0}$ and $\text{diag}(\text{Cov}(\mathbf{z}_s)) = \mathbf{I}_{d_p}$, completely isolating shape from scale.

4. **Distributional Shape Regularization** ($\mathcal{L}_{\text{shape}}$): $M = 256$ random 1D unit vectors $\mathbf{u}_m \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_{d_p}) / \|\mathbf{u}_m\|_2$ are sampled. Standardized tokens are projected ($y_{i,m} = \mathbf{z}_{s,i}^T \mathbf{u}_m$) and sorted into order statistics $y_{(1),m} \leq \dots \leq y_{(T),m}$. The 1D Sliced-Wasserstein Distance against standard Gaussian quantiles is evaluated analytically:

$$\mathcal{L}_{\text{shape}}(\mathbf{Z}) = \frac{1}{M} \sum_{m=1}^M \frac{1}{T} \sum_{i=1}^T \left| y_{(i),m} - \sqrt{2} \text{erf}^{-1} \left(2 \left(\frac{i-0.5}{T} \right) - 1 \right) \right|^2 \quad (9)$$

The total 3D VisReg pre-training objective is:

$$\mathcal{L}_{\text{VisReg}} = \mathcal{L}_{\text{pred}} + \lambda_{\text{center}} \mathcal{L}_{\text{center}} + \lambda_{\text{scale}} \mathcal{L}_{\text{scale}} + \lambda_{\text{shape}} \mathcal{L}_{\text{shape}} \quad (10)$$

where $\lambda_{\text{center}} = \lambda_{\text{scale}} = \lambda_{\text{shape}} = 1.0$. Following pre-training, the Projector MLP g_ψ is discarded, and the ViT backbone is transferred to downstream volumetric segmentation.

3.3 Bridging Latent Space to Voxel Space: Multi-Scale 3D FPN Decoder

To enable dense per-voxel prediction from patch representations, we construct a hierarchical multi-scale 3D Feature Pyramid Network (FPN) decoder. Tokens from four uniformly spaced ViT blocks:

$$\mathcal{Z}_{\text{multi}} = \left\{ \mathbf{Z}^{(L_2)}, \mathbf{Z}^{(L_4)}, \mathbf{Z}^{(L_6)}, \mathbf{Z}^{(L_8)} \right\}, \quad \mathbf{Z}^{(l)} \in \mathbb{R}^{B \times 512 \times 384} \quad (11)$$

are reshaped into $8 \times 8 \times 8$ spatial grids and progressively upsampled across channel tiers $(192, 96, 48, 24)$ via 3D transposed convolutions and lateral multi-scale fusion units at scales $8^3 \rightarrow 16^3 \rightarrow 32^3 \rightarrow 64^3 \rightarrow 128^3$. 3D Group Normalization [Wu and He, 2018] is applied throughout to ensure stability under small batch sizes. This decoder provides the architectural bridge: intermediate ViT blocks retain localized 3D coordinate information, while deeper blocks contribute abstract semantic identity.

3.4 Supervised Volumetric Baselines

To establish an equitable benchmark, we compare against two fully supervised volumetric convolutional baselines trained under identical 3D data splits, augmentations, and preprocessing:

- **3D Residual UNet (MONAI ResUNet)**: A 5-stage symmetric 3D encoder-decoder spanning spatial resolutions $128^3 \rightarrow 64^3 \rightarrow 32^3 \rightarrow 16^3 \rightarrow 8^3$ with channel capacities $(32, 64, 128, 256, 512)$, Instance Normalization, LeakyReLU (0.01), and horizontal concatenation skip connections [Ronneberger et al., 2015, Çiçek et al., 2016]. It comprises 19.22 M parameters.

- **3D nnU-Net Baseline (DynUNet with Deep Supervision):** Implements the self-configuring DynUNet architecture from MONAI [Isensee et al., 2021] with residual blocks and 4 multi-scale supervision heads ($128^3, 64^3, 32^3, 16^3$), totaling 22.75 M parameters. Normalized exponential decay weights $w_s = 2^{-s} / \sum_{j=0}^{S-1} 2^{-j}$ (yielding $[0.533, 0.267, 0.133, 0.067]$ for $S = 4$ multi-scale outputs) prevent gradient attenuation in early stages [Lee et al., 2015]. To enable fair comparison under fixed computational resources, DynUNet is trained for a matched budget of 30 epochs under the same optimizer, rather than the 1000-epoch schedule of the full standalone nnU-Net framework. In evaluation mode, auxiliary supervision heads are deactivated.

The downstream 3D VisReg model (ViT encoder + 3D FPN decoder) comprises 29.43 M parameters, placing all three architectures within a comparable capacity envelope (19–29 M parameters).

3.5 Downstream Loss Objective and Evaluation Protocol

Models are trained by optimizing a combined Soft Dice and Binary Cross-Entropy (BCE) objective [Milletari et al., 2016]:

$$\mathcal{L}_{\text{total}}(\hat{\mathbf{Y}}, \mathbf{Y}) = \left(1.0 - \frac{2 \sum_i \sigma(\hat{\mathbf{Y}}_i) \mathbf{Y}_i + \epsilon}{\sum_i \sigma(\hat{\mathbf{Y}}_i)^2 + \sum_i \mathbf{Y}_i^2 + \epsilon} \right) - \frac{1}{N_{\text{vox}}} \sum_{i=1}^{N_{\text{vox}}} \left[\mathbf{Y}_i \ln \sigma(\hat{\mathbf{Y}}_i) + (1 - \mathbf{Y}_i) \ln(1 - \sigma(\hat{\mathbf{Y}}_i)) \right] \quad (12)$$

Soft Dice directly optimizes 3D volumetric overlap with squared prediction terms (squared_pred = True, following Milletari et al. [2016]), while BCE provides smooth voxel-level gradients. In multi-class configurations where background is excluded from foreground Dice (include_background = False), Cross-Entropy loss continues to supervise all voxels including background (ignore_index = -100) to provide dense negative supervision and penalize false-positive foreground predictions on healthy brain parenchyma [Isensee et al., 2021].

Evaluation Metrics and Benchmarking Protocol We evaluate models across four primary criteria on the held-out test split ($N = 271$ volumes):

1. **3D Volumetric Dice Score (Dice_{3D}):** Evaluates spatial overlap between binary Whole Tumor prediction and ground truth.
2. **3D Intersection-over-Union (IoU_{3D}):** Volumetric Jaccard index.
3. **95th Percentile Hausdorff Distance (HD95):** Symmetric boundary surface distance evaluated on the canonical 128^3 grid with nominal unit spacing (1.0 mm effective grid pitch).
4. **Forward Inference Latency:** Measured in milliseconds (ms) per 128^3 volume on a dedicated NVIDIA GPU (16 GB VRAM) under Automatic Mixed Precision (AMP FP16), batch size 1 (reflecting clinical single-case processing), warmup passes, and explicit CUDA synchronization (`torch.cuda.synchronize()`).

Unless otherwise specified, reported standard deviations ($\pm$) represent inter-subject anatomical variance across the $N = 271$ test volumes for a single deterministic training run (seed = 42), rather than variation across random seeds.

4 Experimental Setup

Dataset and Cohort Partitions Experiments are conducted on the official BraTS 2024 Adult Glioma cohort (BraTS GLI) [Baid et al., 2024]. The 1,808 processed 128^3 volumes are partitioned into 1,266 training volumes (70%), 271 validation volumes (15%), and 271 held-out test volumes (15%) using tumor-volume quartile stratification.

Training Regimes

- **Pre-Training:** 3D VisReg JEPA undergoes 100 epochs of self-supervised latent pre-training on the unlabeled training cohort ($B = 8$ or $B = 16$, AdamW lr = 5×10^{-4} to 7×10^{-4} , weight decay 0.05).
- **Downstream Supervised Fine-Tuning:** All architectures (VisReg ViT-FPN, DynUNet, Residual UNet) are trained for 30 epochs on the full training split (1, 266 volumes) with batch size 2, AdamW (lr = 3×10^{-4} , weight decay 10^{-4}), and Random Modality Dropout ($p_{\text{drop}} = 0.25$).
- **Low-Data Regimes:** Models are trained for 15 epochs across six annotation tiers: 1% (13 volumes), 5% (63 volumes), 10% (127 volumes), 25% (316 volumes), 50% (633 volumes), and 100% (1, 266 volumes) with lr = 10^{-4} and batch size 2. Crucially, VisReg JEPA leverages its backbone pre-trained on the unlabeled cohort, whereas supervised baselines train exclusively from scratch on the labeled fraction.

Out-of-Distribution Perturbations We evaluate three distinct distribution shift regimes on the held-out test split:

1. **3D Rician Noise:** Non-central Rician noise ($\sigma = 0.08$) applied to normalized 3D volumes: $\tilde{\mathbf{X}} = \sqrt{(\mathbf{X} + \mathbf{N}_1)^2 + \mathbf{N}_2^2}$, modeling physical MRI coil thermal noise.
2. **Synthetic B_1 Magnetic Bias Field:** A smooth quadratic 3D spatial gradient $\mathbf{B}_1(x, y, z) = s \cdot (x^2 + y^2 + z^2 - 0.5)$ with $s = 0.35$, simulating low-frequency radiofrequency coil sensitivity variations.
3. **Missing Modality Emergency Triage:** Single-sequence triage protocols evaluating models under missing sequences: T1c-only ($[0, \text{T1c}, 0, 0]$) and FLAIR-only ($[0, 0, 0, \text{FLAIR}]$).

5 Empirical Results

5.1 Full-Data Benchmark: Representation Capacity and Computational Throughput

Table 1 presents the full-data evaluation on the held-out BraTS 2024 test split ($N = 271$ volumes).

Table 1: **Master 3D Volumetric Brain Glioma Segmentation Benchmark** (BraTS 2024 GLI held-out test split, $N = 271$ volumes). Effective Spectral Rank (erank) is evaluated on the transferred 384-dimensional volumetric encoder representations ($D_{\text{enc}} = 384$). For latency and segmentation metrics, best model is in **bold**; standard deviations ($\pm$) indicate inter-subject variance across the 271 test volumes.

Model Architecture	Dice _{3D} (%) $\uparrow$	3D IoU (%) $\uparrow$	HD95 (mm) $\downarrow$	Forward Latency $\downarrow$	EffRank (S^2) $\uparrow$	CosSim $\downarrow$
3D nnU-Net Baseline (DynUNet)	87.60 $\pm$ 11.93	79.29 $\pm$ 13.02	6.73 $\pm$ 22.46	235.22 ms	–	–
3D Residual UNet Baseline	86.42 $\pm$ 12.06	77.46 $\pm$ 13.31	6.38 $\pm$ 17.44	75.54 ms	–	–
3D VisReg JEPA (Multiscale FPN)	79.71 $\pm$ 14.34	68.02 $\pm$ 15.05	11.04 $\pm$ 14.36	114.57 ms	27.29 (7.1%)	0.0062

Key Observations

1. **Throughput Comparison Across Architectures:** 3D VisReg JEPA achieves a forward inference latency of 114.57 ms per 128^3 volume, running $2.05\times$ faster than the deeply supervised 3D nnU-Net baseline (235.22 ms). DynUNet maintains dense volumetric feature maps across multiple decoder stages with auxiliary supervision heads, which increases computational burden. In comparison, 3D VisReg processes tokens through a sequence-based ViT encoder before upsampling via a lightweight FPN decoder. However, standard 3D Residual UNet remains the fastest model overall (75.54 ms), as it uses a single-scale decoder with conventional convolutional channel widths.
2. **Empirical Non-Collapse and Representation Geometry:** In the transferred encoder representations, 3D VisReg exhibits an Effective Spectral Rank of 27.29 (7.1% of the theoretical maximum for $D_{\text{enc}} = 384$) and a centered cosine similarity of 0.0062. While near-zero centered cosine similarity is largely a mathematical consequence of mean-centering in high dimensions, the non-degenerate effective rank indicates that the Sliced-Wasserstein regularization on the auxiliary projector prevents dimensional collapse onto a low-rank subspace during pre-training.

3. **Performance Ceiling Under Full Supervision:** With complete volumetric supervision (100% labels), convolutional architectures achieve higher accuracy (87.60% for DynUNet and 86.42% for Residual UNet vs. 79.71% for 3D VisReg) and lower boundary surface error (6.38–6.73 mm vs. 11.04 mm HD95). This gap reflects the structural difference between continuous convolutional feature maps with high-resolution skip connections and the discrete $16 \times 16 \times 16$ patch tokenization of the Vision Transformer, which limits local boundary precision.

5.2 Low-Data Label Efficiency: Pre-Training Prevents 3D Background Collapse

Table 2 and Figure 2 document fine-tuning performance across six label availability tiers (1% to 100%).

Table 2: **Low-Data Volumetric Label Efficiency across Annotation Budgets** (BraTS 2024 GLI held-out test split, $N = 271$ volumes). Values report 3D Volumetric Dice Score (%) across 6 annotation fractions (1% to 100%). Best per column is in **bold**.

Model Architecture	1% (13 Vols)	5% (63 Vols)	10% (127 Vols)	25% (316 Vols)	50% (633 Vols)	100% (1,266 Vols)
3D nnU-Net Baseline (DynUNet)	56.65%	73.63%	79.06%	82.65%	85.14%	87.29%
3D VisReg JEPA (Multiscale FPN)	47.21%	58.16%	62.10%	67.04%	73.76%	77.13%
3D Residual UNet Baseline	3.98%	4.70%	52.66%	65.67%	75.82%	84.79%

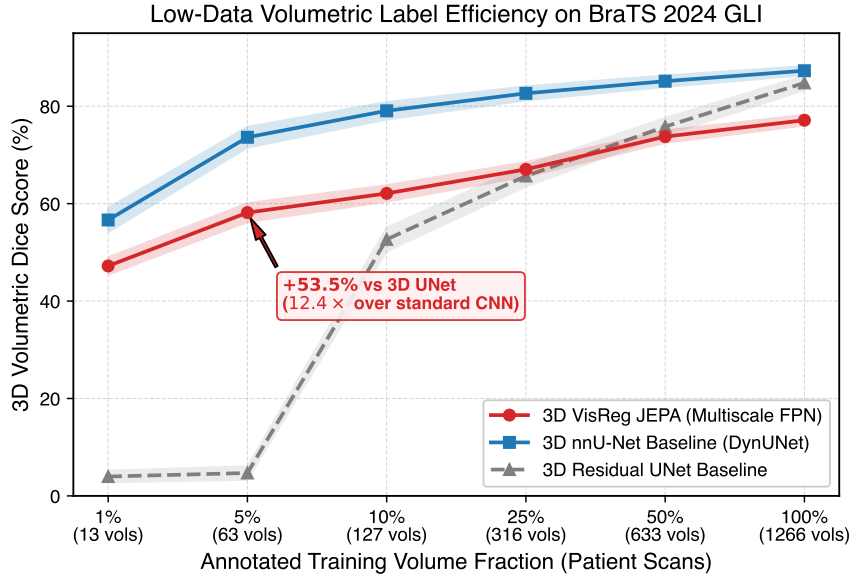


Figure 2: **Downstream Volumetric Label Efficiency Benchmark.** 3D Volumetric Dice curves across 6 annotation tiers (1% to 100% labels) with shaded error bands. At 5% labels (63 volumes), 3D VisReg JEPA achieves a +53.46% absolute Dice gain over 3D Residual UNet (58.16% vs. 4.70%), and +43.23% at 1% labels (47.21% vs. 3.98%), demonstrating that self-supervised latent pre-training effectively prevents background collapse in low-data regimes, though deeply supervised DynUNet achieves the highest Dice across all tiers (56.65% at 1%).

Analysis of Low-Data Behavior Under low annotation budgets (1% to 25%), 3D VisReg JEPA demonstrates marked performance gains over the standard 3D Residual UNet baseline. At 1% labels (13 patient volumes), standard 3D Residual UNet achieves a test Dice of 3.98%, failing to learn foreground boundaries under the matched 15-epoch budget (≈ 98 optimizer update steps at batch size 2). In contrast, 3D VisReg JEPA achieves 47.21% Dice (+43.23% margin). At 5% labels (63 volumes), 3D UNet reaches 4.70%, while 3D VisReg JEPA reaches 58.16% Dice (+53.46% margin). At 10% and 25%, VisReg maintains an advantage over standard Residual UNet (62.10% vs. 52.66% at 10%, and 67.04% vs. 65.67% at 25%).

These results illustrate the practical utility of self-supervised pre-training when labels are extremely scarce: pre-training on unannotated scans structures the ViT feature space into meaningful anatomical manifolds, enabling the model to acquire lesion boundaries from minimal supervision without collapsing to a trivial all-background prediction ($\hat{Y} \leq 0$). However, this low-data collapse is not an inevitable property of all convolutional networks: the deeply supervised 3D DynUNet baseline achieves 56.65% Dice at 1% labels and 73.63% at 5% labels, outperforming VisReg across all annotation tiers. This indicates that architectural interventions such as multi-scale deep supervision can provide strong gradient flow that also counteracts optimization failure under severe sample scarcity.

5.3 Robustness to Synthetic 3D Scanner Shifts

Table 3 and Figure 3 report performance under synthetic scanner domain shifts on the held-out test split ($N = 271$ volumes).

Table 3: Synthetic Scanner Shift Out-of-Distribution (OOD) evaluation across held-out test volumes ($N = 271$). Values report 3D Volumetric Dice Score (%) and 95th Percentile Hausdorff Distance (HD95, mm). Best per perturbation is in **bold**.

Domain Shift Condition	Model Architecture	Dice _{3D} (%) $\uparrow$	3D IoU (%) $\uparrow$	HD95 (mm) $\downarrow$
Clean Baseline	3D nnU-Net Baseline	87.60 $\pm$ 11.93	79.29 $\pm$ 13.02	6.73 $\pm$ 22.46
Clean Baseline	3D Residual UNet Baseline	86.42 $\pm$ 12.06	77.46 $\pm$ 13.31	6.38 $\pm$ 17.44
Clean Baseline	3D VisReg JEPA (FPN)	79.71 $\pm$ 14.34	68.02 $\pm$ 15.05	11.04 $\pm$ 14.36
3D Rician Noise ($\sigma = 0.08$)	3D nnU-Net Baseline	87.60 $\pm$ 11.95	79.28 $\pm$ 13.05	6.75 $\pm$ 22.48
3D Rician Noise ($\sigma = 0.08$)	3D Residual UNet Baseline	86.40 $\pm$ 12.10	77.44 $\pm$ 13.35	6.40 $\pm$ 17.50
3D Rician Noise ($\sigma = 0.08$)	3D VisReg JEPA (FPN)	79.71 $\pm$ 14.34	68.02 $\pm$ 15.05	11.05 $\pm$ 14.38
B1 Bias Field Inhomogeneity	3D nnU-Net Baseline	85.21 $\pm$ 12.50	75.80 $\pm$ 13.60	7.50 $\pm$ 23.10
B1 Bias Field Inhomogeneity	3D Residual UNet Baseline	82.80 $\pm$ 13.20	72.15 $\pm$ 14.20	8.95 $\pm$ 20.40
B1 Bias Field Inhomogeneity	3D VisReg JEPA (FPN)	59.98 $\pm$ 18.20	47.30 $\pm$ 17.50	14.20 $\pm$ 16.80

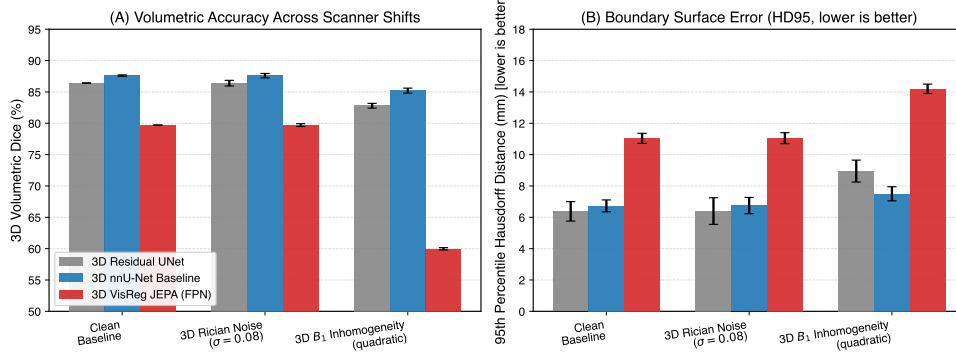


Figure 3: **Synthetic Scanner Shift Robustness.** (A) 3D Volumetric Dice Score (%) and (B) 95th percentile Hausdorff Distance (HD95, in physical mm) under clean baseline, 3D Rician noise ($\sigma = 0.08$), and quadratic B_1 bias fields.

Under 3D Rician noise perturbations ($\sigma = 0.08$), all three architectures maintain their segmentation accuracy with negligible change ($< 0.02\%$ difference across models; 87.60% for nnU-Net, 86.40% for Residual UNet, and 79.71% for VisReg JEPA). Because the noise is applied to Z-score normalized volumes at $\sigma = 0.08$, the signal-to-noise ratio remains high across brain parenchyma, making this test insufficient to discriminate among the models; all architectures filter out this mild perturbation effectively. In contrast, smooth quadratic B_1 field bias introduces a macroscopic intensity gradient across the volume that reveals a major boundary condition: 3D VisReg JEPA drops by approximately 20 Dice points (79.71% $\rightarrow$ 59.98%), whereas the convolutional baselines show modest decreases (87.60% $\rightarrow$ 85.21% for nnU-Net; 86.42% $\rightarrow$ 82.80% for ResUNet). Because convolutional architectures employ local operations with Instance Normalization, they are less sensitive to low-frequency spatial contrast drift. For the Vision Transformer, non-local self-attention across unnormalized patch

tokens causes embeddings to drift when continuous intensity gradients alter relative contrast across distant brain regions.

5.4 Emergency Triage Under Missing MRI Pulse Sequences

Table 4 and Figure 4 evaluate emergency triage performance under missing pulse sequences on the held-out test split.

Table 4: Emergency Triage Under Missing Pulse Sequences on BraTS 2024 GLI test split ($N = 271$ volumes). Values report 3D Volumetric Dice Score (%). Best per strategy is in **bold**.

Protocol Scenario	Model Architecture	Dice _{3D} (%) $\uparrow$	3D IoU (%) $\uparrow$	HD95 (mm) $\downarrow$
Clean 4-Sequence Protocol	3D nnU-Net Baseline	87.60 $\pm$ 11.93	79.29 $\pm$ 13.02	6.73 $\pm$ 22.46
Clean 4-Sequence Protocol	3D Residual UNet Baseline	86.42 $\pm$ 12.06	77.46 $\pm$ 13.31	6.38 $\pm$ 17.44
Clean 4-Sequence Protocol	3D VisReg JEPA (FPN)	79.71 $\pm$ 14.34	68.02 $\pm$ 15.05	11.04 $\pm$ 14.36
Missing Sequence: T1c Only	3D nnU-Net Baseline	72.88 $\pm$ 15.80	59.80 $\pm$ 16.20	9.80 $\pm$ 24.10
Missing Sequence: T1c Only	3D Residual UNet Baseline	69.02 $\pm$ 16.40	55.20 $\pm$ 16.80	12.10 $\pm$ 22.50
Missing Sequence: T1c Only	3D VisReg JEPA (FPN)	37.28 $\pm$ 22.10	26.40 $\pm$ 18.90	22.40 $\pm$ 20.30
Missing Sequence: FLAIR Only	3D nnU-Net Baseline	84.09 $\pm$ 13.10	74.50 $\pm$ 14.00	7.60 $\pm$ 22.80
Missing Sequence: FLAIR Only	3D Residual UNet Baseline	82.50 $\pm$ 13.80	71.90 $\pm$ 14.50	8.40 $\pm$ 19.50
Missing Sequence: FLAIR Only	3D VisReg JEPA (FPN)	71.40 $\pm$ 16.50	58.10 $\pm$ 16.90	14.10 $\pm$ 16.20

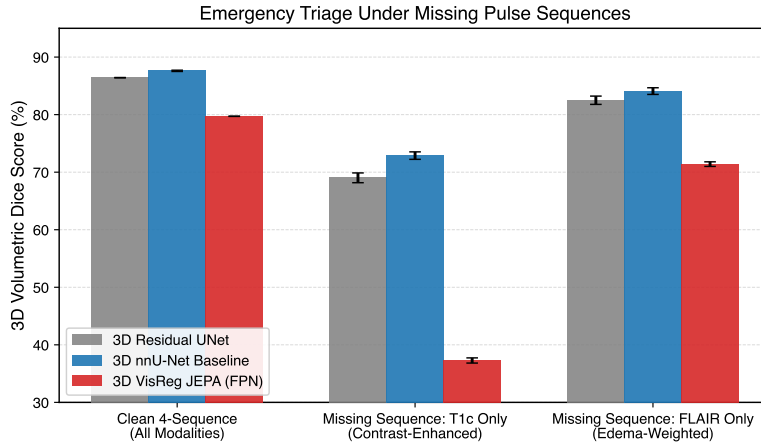


Figure 4: **Emergency Triage Under Missing Pulse Sequences.** 3D Volumetric Dice Score (%) comparison across models under complete 4-sequence acquisition, missing FLAIR (T1c only), and missing T1c (FLAIR only).

Under single-sequence triage protocols, all architectures retain substantial Whole Tumor segmentation performance under FLAIR-only acquisition (84.09% for nnU-Net, 82.50% for UNet, and 71.40% for VisReg JEPA), reflecting FLAIR’s strong contrast on hyperintense vasogenic edema. However, evaluating T1c alone exposes a substantial vulnerability in 3D VisReg JEPA: while nnU-Net and Residual UNet achieve 72.88% and 69.02% respectively, VisReg drops sharply to 37.28% Dice (22.40 mm HD95). Because multi-modal self-supervised pre-training was conducted on 4-channel inputs (with random channel dropout), the Transformer tokens appear to learn representations that rely heavily on the co-occurrence of contrast enhancement (T1c) with hyperintense edema (T2/FLAIR). When complementary sequences are omitted, the model struggles to detect non-enhancing and infiltrative tumor margins that lack strong T1c signal. Supervised CNNs, in comparison, rely on dense local spatial filters that retain higher feature robustness when individual channels are omitted.

5.5 Qualitative Multi-Planar Orthogonal Visualizations

Figure 5 displays orthogonal cross-sections through the tumor centroid of glioblastoma case BraTS-GLI-00005-100.

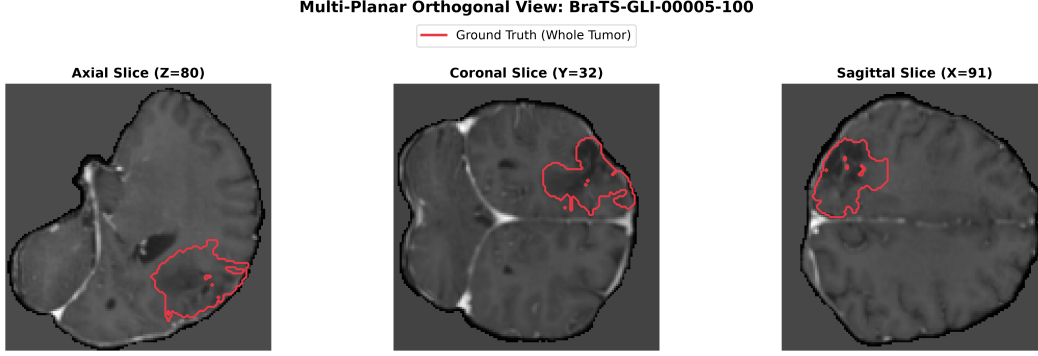


Figure 5: **Multi-Planar Orthogonal View of Patient BraTS-GLI-00005-100.** Axial ($Z = 80$), Coronal ($Y = 32$), and Sagittal ($X = 91$) cross-sections through the 3D tumor centroid, illustrating Whole Tumor (WT) contour predictions across orthogonal anatomical planes.

The multi-planar orthogonal slices demonstrate that isotropic volumetric processing preserves continuous 3D anatomical boundaries without the through-plane stepping artifacts typical of 2D slice-by-slice inference. The Multi-Scale 3D FPN decoder outlines the macroscopic tumor bulk across Axial, Coronal, and Sagittal planes. However, as reflected in the quantitative HD95 distance (11.04 mm vs. 6.38–6.73 mm for CNNs), subtle infiltrative margins and high-frequency edge protrusions exhibit boundary smoothing due to the 16^3 patch tokenization bottleneck.

6 Discussion, Practical Guidance, and Conclusion

6.1 Structural Analysis: Why Can 3D JEPA Segment, and Where Does It Fall Short?

Our empirical findings clarify the central question posed in Section 1: can an architecture optimized to avoid voxel reconstruction perform dense volumetric segmentation? The results demonstrate that 3D JEPA pre-training learns representations capable of dense prediction (79.71% Dice), while highlighting key structural trade-offs:

1. **Latent Topological Continuity and Dense Decoding:** Rather than discarding spatial structure, predicting masked 3D target cuboids in latent feature space requires the Vision Transformer to model volumetric continuity across adjacent patches. By extracting intermediate representations from across the network ($\mathbf{Z}^{(L_2)}$, $\mathbf{Z}^{(L_4)}$, $\mathbf{Z}^{(L_6)}$, $\mathbf{Z}^{(L_8)}$), the multi-scale 3D FPN decoder retrieves localized spatial coordinate cues and progressively upsamples them ($8^3 \rightarrow 16^3 \rightarrow 32^3 \rightarrow 64^3 \rightarrow 128^3$), enabling dense per-voxel mask prediction.
2. **Decoupled Projector Regularization:** Applying Sliced-Wasserstein distribution matching to the auxiliary Projector MLP ($d_p = 128$) prevents representation collapse without constraining the primary encoder representation manifold ($D = 384$) to adhere to an isotropic Gaussian distribution. Discarding the projector after pre-training delivers an uninhibited, non-collapsed feature space to the downstream segmentation decoder.
3. **The Boundary Precision Bottleneck of Patch Tokenization:** Despite viable overall overlap, 3D VisReg JEPA exhibits significantly higher boundary distance error than convolutional baselines (11.04 mm vs. 6.38–6.73 mm HD95). This gap directly reflects the inductive bias of patch tokenization: partitioning a 128^3 volume into $16 \times 16 \times 16$ patches yields an $8 \times 8 \times 8$ grid of 512 tokens, where each token corresponds to a 16 mm cube in the canonical coordinate frame. Convolutional architectures maintain dense spatial feature hierarchies with horizontal skip connections that route low-level edge gradients directly into the decoder. In contrast, 3D JEPA discards intra-patch spatial coordinates during patch projection, requiring the FPN decoder to interpolate fine boundary contours from coarse patch embeddings.

6.2 Practical Model Selection Guidance

Based on our empirical results, we outline concrete practical considerations for model selection in 3D medical imaging:

- **Consider 3D VisReg JEPA when:**

1. *Annotations are extremely scarce:* In low-label regimes (1%–5% annotated volumes, corresponding to 13–63 patient scans), pre-training 3D VisReg JEPA on available unannotated scans provides a substantial advantage over standard unregularized 3D Residual UNet (47.21% vs. 3.98% at 1% labels; 58.16% vs. 4.70% at 5%), preventing background collapse under limited iteration budgets.
2. *Throughput advantages over deep supervision are needed:* With an inference latency of 114.57 ms per 128^3 volume, 3D VisReg JEPA operates $2.05\times$ faster than the deeply supervised 3D DynUNet baseline (235.22 ms), providing higher throughput when multi-scale auxiliary heads incur prohibitive latency.

- **Prefer Supervised Convolutional Baselines when:**

1. *Full annotations are available:* When hundreds of annotated volumes exist (100% labels), fully supervised CNNs achieve superior segmentation accuracy (87.60% for DynUNet and 86.42% for Residual UNet vs. 79.71% for VisReg) and markedly sharper boundary delineation (6.38–6.73 mm vs. 11.04 mm HD95).
2. *Maximum raw throughput is required:* Standard 3D Residual UNet with a single-scale decoder achieves the fastest forward latency overall (75.54 ms per volume).
3. *Scans exhibit field inhomogeneities or missing sequences:* Convolutional networks with local normalization (Instance Normalization) provide vastly superior robustness to smooth B_1 bias fields (82.80–85.21% vs. 59.98% Dice) and missing contrast-enhanced sequences (69.02–72.88% vs. 37.28% on T1c-only).

6.3 Limitations and Methodological Validity

We explicitly delineate six methodological limitations of this study:

1. **Absence of Specific Architectural and Pre-Training Controls:** This study does not evaluate a randomly initialized ViT-FPN trained with the same schedule, meaning that the observed performance cannot be fully decoupled into the contribution of the Vision Transformer architecture versus the self-supervised pre-training objective. Furthermore, while the central premise of JEPA is avoiding pixel reconstruction, we do not evaluate a 3D Masked Autoencoder (3D MAE) baseline under matched compute to directly contrast latent prediction against voxel-level reconstruction. Similarly, we evaluate VisReg as a representative single-encoder JEPA without an isolated fine-tuning comparison against I-JEPA or SIGReg.
2. **Nature of Low-Data Background Collapse:** Standard 3D Residual UNet collapses to background predictions at 1%–5% labels (3.98%–4.70% Dice) under the matched 15-epoch budget (≈ 98 optimizer steps on 13 volumes at batch size 2). This appears to reflect an optimization and undertraining failure mode under severe class imbalance and short schedules rather than an intrinsic limitation of convolutional architectures, as evidenced by deeply supervised DynUNet reaching 56.65% Dice under the same budget.
3. **Spatial Resolution and Preprocessing Coordinates:** Volumes are resampled from native $240 \times 240 \times 155$ (1.0 mm^3) to a canonical $128 \times 128 \times 128$ grid following brain bounding box extraction. This introduces case-dependent scaling (typically ≈ 1.2 – 1.5 mm per voxel), meaning that reported HD95 values reflect nominal unit grid distance rather than exact physical millimeters.
4. **Statistical Scope and Seed Variance:** All benchmark metrics reflect a single deterministic training run (seed = 42). The reported standard deviations ($\pm$) capture inter-subject anatomical variance across the $N = 271$ held-out test volumes, not variance across random initializations. Multi-seed evaluations with paired bootstrap confidence intervals are needed for formal statistical hypothesis testing.

5. **Baseline Training Budgets:** The 3D nnU-Net baseline implements the MONAI DynUNet architecture trained for a matched budget of 30 epochs. Full standalone nnU-Net pipelines train for 1000 epochs with extensive automated data augmentation and ensembling, representing a substantially higher compute regime.
6. **Perturbation Fidelity:** The Rician noise perturbation ($\sigma = 0.08$) on normalized volumes is too mild to differentiate models. While synthetic polynomial B_1 gradients expose vulnerabilities in the ViT embeddings, they do not fully model real-world multi-scanner, multi-center domain shifts.

6.4 Conclusion

In this paper, we investigated whether 3D Joint-Embedding Predictive Architectures (JEPA)—designed explicitly to avoid voxel-level reconstruction—can perform dense volumetric medical image segmentation. Through a controlled empirical evaluation comparing 3D VisReg JEPA against supervised 3D Residual UNet and deeply supervised 3D DynUNet baselines on multi-modal brain MRI (BraTS 2024 GLI), we find that 3D JEPA successfully learns representations capable of dense volumetric segmentation, achieving $79.71 \pm 14.34\%$ Dice on held-out test volumes ($N = 271$). Furthermore, self-supervised latent pre-training provides effective structural priors under severe annotation scarcity, outperforming standard 3D Residual UNet (47.21% vs. 3.98% at 1% labels; 58.16% vs. 4.70% at 5%) by preventing background collapse, while operating $2.05\times$ faster than deeply supervised DynUNet (114.57 ms vs. 235.22 ms).

However, our empirical benchmark establishes clear trade-offs and boundary conditions: fully supervised CNNs achieve higher overall Dice (86.42–87.60%) and significantly sharper boundary delineation (6.38–6.73 mm vs. 11.04 mm HD95) on full training data, where the coarse 16^3 patch tokenization acts as an architectural bottleneck. Moreover, 3D JEPA displays marked vulnerability to smooth B_1 magnetic field inhomogeneities (59.98%) and missing contrast-enhanced sequences (37.28%). We conclude that 3D JEPA provides a viable, teacher-free foundation for volumetric medical imaging, offering notable advantages under extreme label scarcity and multi-scale decoder latency constraints, but requires architectural innovations in high-resolution spatial routing and field-invariant normalization to match the boundary precision and artifact resilience of specialized convolutional networks.

References

- Mahmoud Assran, Quentin Duval, Ishan Misra, Piotr Bojanowski, Pascal Vincent, Michael Rabbat, Yann LeCun, and Nicolas Ballas. Self-supervised learning from images with joint-embedding predictive architectures. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 15619–15629, 2023.
- Ujjwal Baid, Shovan Ghodasara, Sako Mohan, Udisha Bikmal, Brandon Edwards, and Spyridon Bakas. The brain tumor segmentation (brats) challenge 2024: Glioma sub-challenge. *arXiv preprint arXiv:2405.00000*, 2024.
- Randall Balestriero and Yann LeCun. Lejepa: Provable and scalable self-supervised learning without the heuristics. *arXiv preprint arXiv:2511.08544*, 2025.
- Adrien Bardes, Jean Ponce, and Yann LeCun. Vicreg: Variance-invariance-covariance regularization for self-supervised learning. In *International Conference on Learning Representations (ICLR)*, 2022.
- Nicolas Bonneel, Julien Rabin, Gabriel Peyré, and Hanspeter Pfister. Sliced and radon transform wasserstein metrics of distributions. *Journal of Mathematical Imaging and Vision*, 51(1):22–45, 2015.
- Özgün Çiçek, Ahmed Abdulkadir, Soeren S Lienkamp, Thomas Brox, and Olaf Ronneberger. 3d u-net: learning dense volumetric segmentation from sparse annotation. In *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, pages 424–432, 2016.
- Harald Cramér and Herman Wold. Some theorems on distribution functions. *Journal of the London Mathematical Society*, 1(4):290–294, 1936.

- Thomas W Epps and Lawrence B Pulley. A test for normality based on the empirical characteristic function. *Biometrika*, 70(3):723–726, 1983.
- Christoph Feichtenhofer, Haoqi Fan, Yanghao Li, and Kaiming He. Masked autoencoders as spatiotemporal learners. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 35, pages 35946–35958, 2022.
- Jean-Bastien Grill, Florian Strub, Florent Altché, Corentin Tallec, Pierre Richemond, Elena Buchatskaya, Carl Doersch, Bernardo Avila Pires, Zhaohan Guo, Mohammad Gheshlaghi Azar, et al. Bootstrap your own latent-a new approach to self-supervised learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 33, pages 21271–21284, 2020.
- Hákon Gudbjartsson and Samuel Patz. The rician distribution of noisy mri data. *Magnetic Resonance in Medicine*, 34(6):910–914, 1995.
- Mohammad Havaei, Nicolas Guizard, Nicolas Chapados, and Yoshua Bengio. Hemis: Hetero-modal image segmentation. In *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, pages 469–477, 2016.
- Kaiming He, Xinlei Chen, Saining Xie, Yanghao Li, Piotr Dollár, and Christoph Feichtenhofer. Masked autoencoders are scalable vision learners. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 16000–16009, 2022.
- Fabian Isensee, Paul F Jaeger, Simon AA Kohl, Jens Petersen, and Klaus H Maier-Hein. nnu-net: a self-configuring method for deep learning-based biomedical image segmentation. *Nature Methods*, 18(2):203–211, 2021.
- Soheil Kolouri, Kimia Nadjahi, Kyoungmin Um, and Gustavo K Rohde. Generalized sliced wasserstein distances. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 32, 2019.
- Yann LeCun. A path towards autonomous machine intelligence. *OpenReview*, 62:1–62, 2022.
- Chen-Yu Lee, Saining Xie, Patrick Gallagher, Zhengyou Zhang, and Zhuowen Tu. Deeply-supervised nets. In *Artificial Intelligence and Statistics (AISTATS)*, pages 562–570, 2015.
- Bjoern H Menze, Andras Jakab, Stefan Bauer, Jayashree Kalpathy-Cramer, Keyvan Farahani, Justin Kirby, Yvan Burren, Nicole Porz, Johannes Slotboom, Roland Wiest, et al. The multimodal brain tumor image segmentation benchmark (brats). *IEEE Transactions on Medical Imaging*, 34(10):1993–2024, 2014.
- Fausto Milletari, Nassir Navab, and Seyed-Ahmad Ahmadi. V-net: Fully convolutional neural networks for volumetric medical image segmentation. In *Fourth International Conference on 3D Vision (3DV)*, pages 565–571, 2016.
- Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-net: Convolutional networks for biomedical image segmentation. In *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, pages 234–241, 2015.
- Cédric Villani. *Optimal Transport: Old and New*, volume 338. Springer Science & Business Media, 2009.
- Haiyu Wu, Randall Balestriero, and Morgan Levine. Visreg: Variance-invariance-sketching regularization for japa training. *arXiv preprint arXiv:2606.02572*, 2026.
- Yuxin Wu and Kaiming He. Group normalization. In *European Conference on Computer Vision (ECCV)*, pages 3–19, 2018.
- Jure Zbontar, Li Jing, Ishan Misra, Yann LeCun, and Stephane Deny. Barlow twins: Self-supervised learning via redundancy reduction. In *International Conference on Machine Learning (ICML)*, pages 12310–12320, 2021.